

Tree-Level-Improved Symanzik Gluon Action

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To improve the Wilson gauge action at $\mathcal{O}(a^2)$ requires at least three dimension-six lattice operators. The simplest are the rectangle, parallelogram, and chair (sometimes called a bent rectangle). These are depicted in Fig. 1 of Ref. [1] and denoted with integers 1, 2, and 3, respectively. An improvement on the Wilson action can then be written as

$$S_{\text{imp}} = \frac{2}{g_0^2} \sum_x \sum_{i=0}^3 c_i \text{Re tr} (\mathbf{1} - U_i(x)), \quad (1)$$

with $i = 0$ indicating the plaquette, and U_i being the corresponding lattice operator (summed over all orientations). The common choice

$$c_0 = \frac{5}{3}, \quad c_1 = -\frac{1}{12} \quad (2)$$

enforces at tree-level the absence of $\mathcal{O}(a^2)$ lattice artifacts in the static-quark potential. For more information, including other possible choices of coefficients, see Ref. [1].

References

- [1] M. Lüscher and P. Weisz. On-shell improved lattice gauge theories. *Comm. Math. Phys.*, 97(1):59–77, 1985.